

ECE 315: Signals and Systems

Homework 1

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This homework is due on **2/6, 11:59pm**. For analytical problems, write your solutions independently by hand. For coding problems, attach your source code and answers. Combine them in one PDF document and submit on CANVAS. Please ensure that your PDF is clearly readable on a computer.

1. (4pts) Consider the following continuous-time signals:

- $x(t) = e^{-t}$, $t \geq 0$
- $x(t) = \cos(\frac{\pi}{2}t)$
- $x(t) = \sin(\frac{\pi}{2}t) \cos(\frac{\pi}{2}t)$
- $x(t) = e^{-j\frac{\pi}{2}t} + e^{-j\frac{\pi}{3}t}$

(i) Compute their energy and average power.

(ii) Determine whether they are periodic and if so, compute their fundamental periods.

(iii) Repeat (i) and (ii) by changing the index time t to n , i.e., $x[n]$.

2. (1pts) Consider a real-valued signal $x(t) = \begin{cases} t-1, & t \geq 1 \\ 1-t^2, & 0 \leq t < 1 \\ 1, & t < 0. \end{cases}$ Sketch

the graphs of signals $x(t)$, $x(t-1)$, $x(2t)$, and $x(2t-1)$ in the rectangular plane.

3. (2pts) Consider two continuous-time signals $x_1(t)$ and $x_2(t)$ with periods T_1 and T_2 , respectively. Determine the conditions on T_1 and T_2 such that $x_1(t) + x_2(t)$ is periodic. Determine the fundamental period when it is periodic.

4. (2pts) Consider a continuous-time signal $x(t)$. Find its ODD and EVEN parts. Express the energy of $x(t)$ in terms of the energies of the ODD and EVEN parts.

5. (1pts) Consider a continuous-time signal $x(t) = \delta(t+1) - 2\delta(t-2)$. Compute the energy and average power of the signal $y(t) = \int_{-\infty}^t x(\tau)d\tau$.
6. Complete the MATLAB problems **1.11-1**, **1.11-2**, **1.11-3**, **1.11-4** in **LSS** (Page 149).